Meridian Motors Knowledge Graph Dataset

NEXUS Multi-System Integration Test Data — Complete Documentation

- 17 Plants across 6 Countries
- 9 Independent Source Systems
- 3 Vehicle Platforms
- ~121 CSV Files
- 5 Companies

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1 Purpose & Design Philosophy

The Core Problem

Real automotive manufacturing networks do not run on a single system. They operate across **nine independently deployed systems**, built by different vendors across different decades, with no native interoperability. The same physical machine receives a different identifier in each system. The same production line carries different names depending on which system configured it. Some entities known to one system are entirely invisible to another.

The **NEXUS knowledge graph** is designed to reconcile these nine partial, disagreeing views into one coherent picture. To validate this stitching logic, the test data must *disagree with itself* in the same realistic ways that real enterprise systems do.

> **Key Design Principle**
>
> Every CSV file is a deliberately imperfect view of one single underlying reality. The imperfections are not random noise — they are the specific, well-known categories of disagreement that real enterprise systems exhibit.

Three-Stage Construction

1. **Ground Truth First:** A single, perfectly consistent reality was generated — actual plants, lines, machines, and processes with one correct name and one correct ID for every entity. This layer is never exposed; it serves as the internal answer key.
2. **Fracture Into Nine Views:** This single reality was fractured into nine separate exports, each mimicking a real-world system (ERP, MES, QMS, etc.).
3. **Independent Corruption:** Each export was independently altered with realistic data quality issues: different field names, different ID formats, missing fields, typos, inconsistent abbreviations, and blank cells.

Figure 1.1 — Data Generation Pipeline

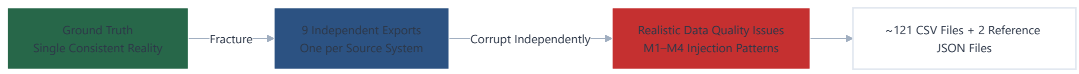

Figure 1.1 — Three-stage construction: a single ground truth is fractured into nine independent exports, each corrupted with realistic data-quality issues.

17 Manufacturing Plants	**6** Countries	**3** Vehicle Platforms
5 Companies	**9** Source Systems	**~121** CSV Files

2 The Meridian Motors Network

Company Ownership Structure

A real OEM manufacturing network consists predominantly of supplier-owned factories, not just captive plants. The Meridian Motors network reflects this reality: 13 plants are OEM-owned, while 4 belong to independent suppliers at Tier-1 and Tier-2 levels.

COMPANY	ROLE	PLANT COUNT	TIER
Meridian Motors	OEM — Final Assembly, Powertrain, Body Manufacturing	13	OEM
Corvus Interiors	Seats & Interior Systems	1	TIER-1
Halberd Systems	Electronics & Wiring Harness	1	TIER-1
Solara Metals	Raw Steel Coil	1	TIER-2
Ferrovia Components	Raw Wiring-Harness Components	1	TIER-2

Table 2.1 — Company roster and ownership tiers within the Meridian Motors manufacturing network.

Plant Classification & Function

CATEGORY	PLANTS	COUNT	FUNCTION
Mother Plants	Engine Plant DE, Engine Plant US, Battery & eDrive Plant PL, Battery & eDrive Plant CN	4	Produce the powertrain "heart" — internal-combustion engines or battery/eDrive systems
Body Manufacturing Chain	Stamping Plant DE → BIW/Welding Plant DE → Paint Shop DE	3	Sequential process chain. Every vehicle platform passes through all three plants.
Tier-1 Component Plants	Transmission Plant PL, Seating & Interior Plant PL (Corvus), Electronics & Wiring Harness Plant MX (Halberd)	3	Produce major sub-assemblies feeding final assembly
Tier-2 Raw Material Plants	Solara Metals Coil Plant (DE), Ferrovia Harness Components Plant (VN)	2	Smallest, least-instrumented plants. Feed raw materials into Tier-1 suppliers.
Final Assembly Plants	Assembly Plant DE, IN, MX, US, CN	5	Finished vehicles roll out here. Platform assignment varies by plant.

Table 2.2 — Plant classification by manufacturing function and position in the supply chain.

Material Flow & Supply Chain Topology

The network is not a flat fan-out. It consists of two genuinely separate supply chains (EV vs. ICE) bridged by three shared universal suppliers. This structural design creates deliberate chokepoints.

Platform Assignment Logic

- **EV only:** Assembly Plant DE and Assembly Plant CN build the EV platform, fed by the two Battery & eDrive plants.
- **ICE only:** Assembly Plant IN, MX, and US build the Compact and SUV platforms, fed by the Engine and Transmission plants.
- **Universal suppliers:** Paint Shop DE, Seating & Interior Plant PL, and Electronics & Wiring Harness Plant MX feed all five assembly plants regardless of platform.

Figure 2.1 — Meridian Motors Supply Chain Network

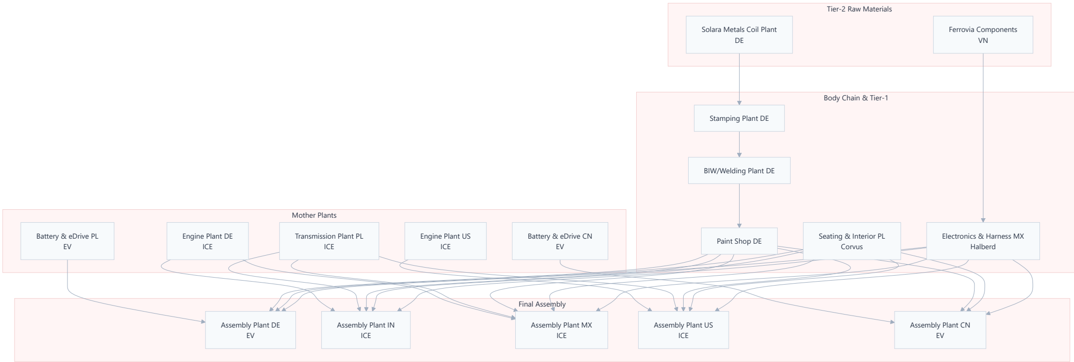

Figure 2.1 — Complete supply chain topology. Two platform-specific lanes (EV in green, ICE in orange) converge at five final assembly plants, bridged by three universal suppliers (purple) and the body manufacturing chain (blue). Tier-2 raw material plants (gray) feed into Tier-1 and body chain entry points.

Structural Chokepoint Analysis

The three bridging plants (Paint Shop DE, Seating & Interior PL, Electronics & Harness MX) plus the body manufacturing chain are the deliberate structural chokepoints of the network. If any single plant were to go offline, the impact traces directly to this topology. A failure at Paint Shop DE, for example, halts all vehicle production across all five assembly plants and all three platforms.

3 The Nine Source Systems

Every plant (with two exceptions) exports data from seven systems. Two additional systems operate at the network level, producing files that cover the entire company rather than a single plant.

#	SYSTEM	REAL-WORLD EQUIVALENT	WHAT IT CAPTURES	SCOPE
1	**ERP**	SAP	Asset register — machines, costs, manufacturers	Per-plant
2	**MES**	ISA-95-style shop-floor system	Physical execution — lines, stations, work units, processes	Per-plant
3	**Master Data (MD)**	Governance spreadsheet	"Official" canonical names for plants and lines	Per-plant
4	**QMS**	IATF 16949 quality system	Defects — non-conformance reports, failure modes, dispositions	Per-plant
5	**EAM**	IBM Maximo / SAP PM	Maintenance — work orders, breakdowns, preventive/corrective/emergency	Per-plant
6	**WMS**	Manhattan / SAP EWM	Material movements — shipments between plants with quantities	Per-plant
7	**SPC**	Minitab / Q-DAS	Statistical process control — measurements against spec limits, Cpk	Per-plant (absent at 2 Tier-2 plants)
8	**PLM**	Siemens Teamcenter / PTC Windchill	Product structure — Bill of Materials per vehicle platform	Network-level (3 files)
9	**APS/SCM**	Kinaxis / SAP IBP	Production planning — weekly capacity, planned volumes, lead times	Network-level (1 file)

Table 3.1 — Complete roster of the nine source systems, their real-world counterparts, and their scope.

Per-Plant vs. Network-Level Scope

The distinction between per-plant and network-level systems is not arbitrary — it mirrors actual enterprise software deployment patterns.

Per-Plant Systems (7)

ERP, MES, MD, QMS, EAM, WMS, and SPC all track physically local phenomena — machines in a specific building, lines running at a specific site, defects found locally, repairs performed locally. Even when every plant nominally runs the same ERP vendor, each instance is configured and maintained independently. This is especially true for the four supplier-owned plants (Corvus, Halberd, Solara, Ferrovia), each running its own IT infrastructure.

Result: 15 plants × 7 systems = 105 files, each with its own quirks.

Network-Level Systems (2)

PLM and APS/SCM are inherently cross-plant. A vehicle's Bill of Materials is engineered once, centrally, by the product engineering organization — it is identical whether the vehicle is built in Mexico or the United States. Capacity planning is cross-plant by definition: Engine Plant DE's output cannot be planned without knowing Assembly IN/MX/US demand.

Result: 3 PLM files (one per platform) + 1 APS/SCM file = 4 network-level files.

Critical Independence Assumption

Each of the nine exports was generated independently. The ERP export has no knowledge of the MES export, and vice versa. This independence is a foundational design choice: it replicates the exact siloed reality of enterprise systems that were never designed to interoperate.

4 File Layout on Disk

The on-disk folder structure follows a strict naming convention. Every plant folder uses the pattern `plant_<function>_<country>`, making it straightforward to locate any file.

Figure 4.1 — Directory Tree Structure

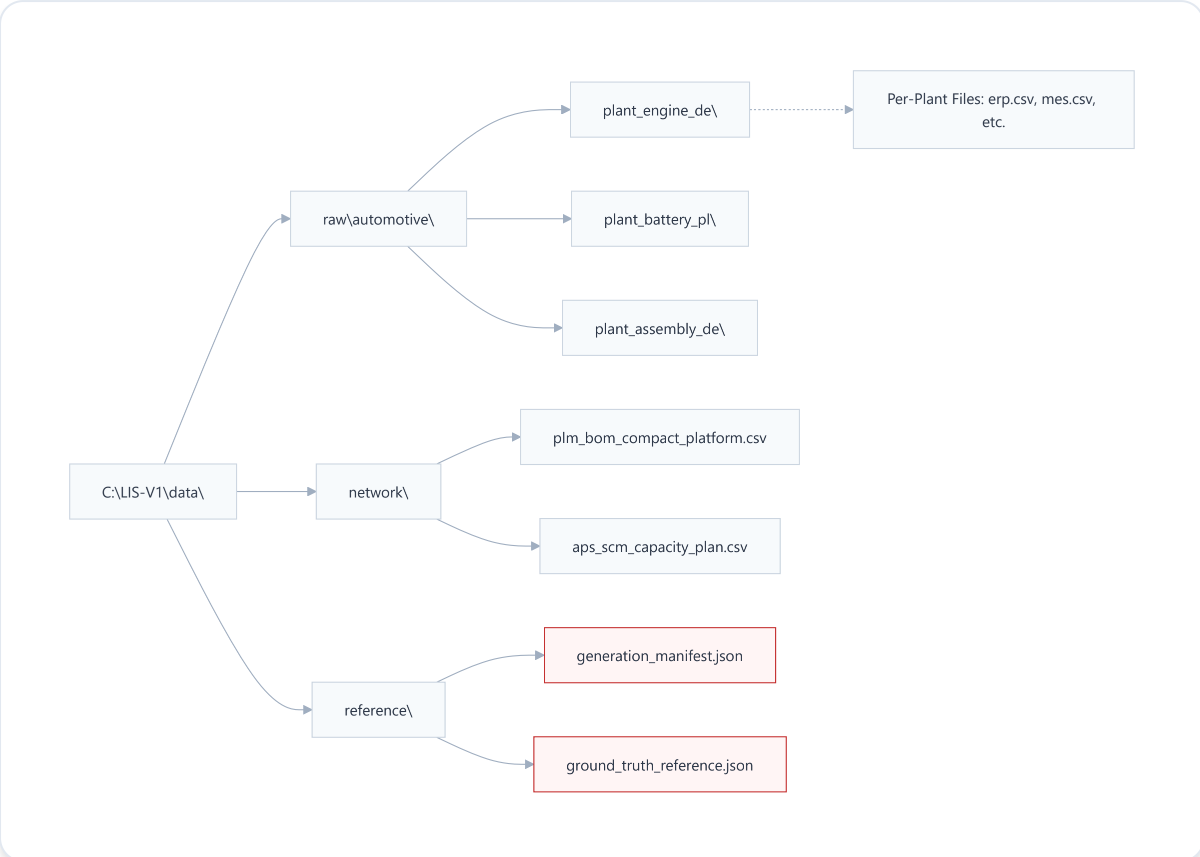

Figure 4.1 — Complete directory tree. Fifteen per-plant folders (each containing 6 or 7 CSVs), one network folder (4 CSVs), and one internal reference folder (2 JSON files). Tier-2 plants lack SPC files by design.

Reference Folder Isolation

The `reference\` folder contains `generation_manifest.json` and `ground_truth_reference.json` — the complete answer key. Exposing these files during any validation exercise defeats the entire purpose of the test, because the objective is to determine whether downstream mapping logic can resolve the mess without prior knowledge of the ground truth. Keep `raw\automotive\` and `reference\` conceptually separate: one is the demo data, the other is the grading key.

5 ERP — Asset Register

The ERP file represents the SAP-style physical asset register. If a machine physically exists on the factory floor, it should have a row here. The ERP department's responsibility is to track what was purchased, what it cost, and who manufactured it.

Field Mapping Across Schema Variants

Three completely different column layouts exist across the network. One layout even omits the cost-center field entirely — a realistic scenario when different sites configure the same ERP vendor independently.

CONCEPT	VARIANT 1 (PAINT SHOP DE)	VARIANT 2 (ENGINE PLANT DE)	VARIANT 3 (BIW PLANT DE)
Plant identifier	PlantCode	Werk	Site_ID
Asset number	FixedAssetNo	Anlagennummer	Asset_Tag
Description	AssetDesc	Anlagenbezeichnung	Equipment_Description
Cost center	*absent*	Kostenstelle	CostCtr
Acquisition value	AcqValue	Anschaffungswert	Book_Value_USD
Manufacturer	Manufacturer	Hersteller	OEM_Name

Table 5.1 — ERP field names across three schema variants. The same semantic concept appears under different column names, and one variant drops the cost-center field entirely.

Sample Data — Variant 1 (Paint Shop DE)

```
PlantCode,FixedAssetNo,AssetDesc,AcqValue,Manufacturer P1060,000070000001,Adhesive Dispensing Unit,35820.11,AIDA Engineering P1060,000070000002,Conveyor System,57731.42,Carl Zeiss Industrial Metrology P1060,000070000003,Palletizing Robot,133040.03,KUKA
```

Sample Data — Variant 2 (Engine Plant DE)

```
Werk,Anlagennummer,Anlagenbezeichnung,Kostenstelle,Anschaffungswert,Hersteller
P1000,000010000001,Torque Application Unit,CC-4645,22683.23,DMG MORI
P1000,000010000002,Battery Pack Assembly Cell,CC-4646,570982.81,Mazak Corporation
P1000,000010000003,Battery Pack Assembly Cell,CC-4646,415304.51,Mazak Corporation
```

Sample Data — Variant 3 (BIW/Welding Plant DE)

```
Site_ID,Asset_Tag,Equipment_Description,CostCtr,Book_Value_USD,OEM_Name
P1050,000060000001,Vision Inspection Station,CC-4891,35775.30,FANUC Corporation
P1050,000060000002,Conveyor System,CC-4891,58743.57,Durr AG
P1050,000060000003,Assembly Torque Wrench Station,CC-4893,21680.96,Haas Automation
```

Deliberate Data Quality Issues

Missing Fields

Approximately 3% of non-critical fields are blank per export. The following rows show blank description fields:

```
P1060,000070000004,,306212.06,kuka ag P1060,000070000008,,150379.01,KUKA
```

Spelling Drift

The same manufacturer appears under multiple spellings even within a single file. KUKA is rendered as `kuka ag` (lowercase) and `KUKA` (abbreviated) in the same export. Across the network, KUKA appears as both `KUKA AG` and `KUKA` in roughly a 2:1 mix — the kind of vendor-master spelling drift that accumulates over a decade of different users entering data.

Cross-System Identity Fragmentation

The asset number in ERP (e.g., `000070000001`) has no relationship to how any other system refers to the same physical machine. MES does not know this number, and EAM will use a

> completely different format (`ASSET-00001`) for the identical machine. This is not an oversight — it is the core reconciliation problem the dataset is built to demonstrate.

6 MES — Shop Floor System

Where ERP tracks assets, MES tracks execution — which line, which station, and which process is running where. This is the system where deliberate identity confusion becomes structurally interesting, not merely cosmetic.

Field Mapping Across Schema Variants

CONCEPT	VARIANT A (ENGINE DE)	VARIANT B (PAINT DE)	VARIANT C (ELECTRONICS MX)
Line	LineID	Line_No	Line_No
Process number	ProcessNo	Process_ID	Process_ID
Process name	ProcessName	Local_Process_Desc	Local_Process_Desc
Work center	WorkCenterID	WC_ID	WC_ID
Station	StationID	Station_No	Station_No

Table 6.1 — MES field names across three schema variants. Note that Variants B and C share column names but contain data from entirely different plants.

Sample Data — Variant A (Engine Plant DE)

```
LineID,ProcessNo,ProcessName,WorkCenterID,StationID L01,PR-0054,Rear Axle
Inspection,WU1,S01 L01,PR-0001,Bead Application,WU2,S01 L01,PR-0076,Fuel Rail
Coating,WU3,S02
```

Sample Data — Variant B (Paint Shop DE)

```
Line_No,Process_ID,Local_Process_Desc,WC_ID,Station_No L01,PR-0059,FLOOR PAN
MOLDING,WU1,S01 L01,PR-0050,ASSEMBLY - TRUNK LID,WU2,S01 L01,PR-0004,AOI
CHECK,WU3,S01
```

M1: The Homonym Problem — Same Label, Different Reality

Every plant's MES numbers its own lines starting from 1 — `L01`, `L02`, `L03` — with zero awareness that another plant is doing the exact same thing. The label `L01` is therefore meaningless in isolation; it only resolves to a specific physical line once the plant context is known.

```
# Engine Plant DE L01, PR-0054, Rear Axle Inspection, WU1, S01 # Electronics & Wiring Harness Plant MX L01, PR-0001, Bead Application, WU1, S01
```

Across the dataset, this collision occurs for **136 plant pairs** — it is not a rare edge case, it is the default state. The same label genuinely refers to different physical things depending on which plant's file you are reading.

M2: The Synonym Problem — Same Reality, Different Label

The process number (`ProcessNo` / `Process_ID`) is the one honest, stable identifier in the entire file; it is globally consistent. However, the human-readable name each plant assigns to that process drifts independently.

```
# Same canonical process PR-0001 at two different plants: Engine Plant DE: PR-0001, "Bead Application" Paint Shop DE: PR-0001, "DISPENSE SEALANT"
```

Both names describe the same underlying adhesive-dispensing operation, but each was configured by different people at different sites. Additionally, Paint Shop DE's entire file renders process names in UPPERCASE — an independent, per-plant styling quirk. Approximately **69%** of all processes in this dataset exhibit this kind of naming drift somewhere in the network, calibrated against real industry research.

> **Analytical Implication**
>
> The question "how would you know Line L01 at two plants are not the same line?" is precisely the right analytical question. The answer is that you cannot determine this from a single file in isolation — the plant-plus-line combination is required. This is exactly the kind of resolution challenge the downstream knowledge graph is designed to solve.

7 Master Data — Governance Record

Master Data is the "official" governance record — a spreadsheet maintained as the single source of truth for plant and line names. In practice, this is the system that gets updated least often, because unlike ERP or MES, nobody's daily workflow depends on it being current. This dataset models that decay deliberately.

Field Mapping Across Schema Variants

CONCEPT	VARIANT A (ENGINE DE)	VARIANT B (SEATING PL)	VARIANT C (STAMPING DE)
Plant name	Facility	Facility	PlantName
Line name	Line_Ref	Line_Ref	LineName
Last updated by	*absent*	*absent*	LastUpdatedBy
Last updated date	Updated_On	Updated_On	LastUpdatedDate

Table 7.1 — Master Data field names across three schema variants. Variant C is the only one that captures who last updated the record.

Sample Data — Variant A (Engine Plant DE)

```
Facility,Line_Ref,Updated_On Engine Plant DE (Old),Engine Plant DE Line 1,2023-05-14
Engine Plant DE (Old),Engine Plant DE Line 2,2021-11-26 Engine Plant DE (Old),Engine
Plant DE Line 3, Engine Plant DE (Old),Engine Plant DE Line 4,2019-05-15 Engine Plant
DE (Old),Engine Plant DE Line 5,2020-10-18
```

Sample Data — Variant B (Seating & Interior Plant PL)

```
Facility,Line_Ref,Updated_On Seating & Interior Plant PL,L01,2020-08-23 Seating &
Interior Plant PL,seating & interior plant pl line 2,2019-05-02 Seating & Interior
Plant PL,seating & interior plant pl line 3, Seating & Interior Plant PL,seating &
interior plant pl line 4,2022-11-18 Seating & Interior Plant PL,seating & interior
plant pl line 5,
```

Three Patterns of Decay

PATTERN 1: STALENESS

Stale Plant Names

The plant name itself is outdated — "Engine Plant DE (Old)" — even though ERP and MES were updated when the plant was renamed. Roughly 1 in 6 plants exhibits this kind of staleness in the MD file.

PATTERN 2: MISSING METADATA

Blank Update Dates

The update date is simply missing on some rows — nobody logged when the row was last touched. The missing-update rate in MD is approximately 20%, noticeably worse than ERP's ~3% blank rate, reinforcing that MD genuinely is the messiest of the three classic systems.

PATTERN 3: IDENTITY COLLAPSE

Line ID Instead of Name

The line name field sometimes contains only the local line ID ("L01") instead of a human-readable name, or is completely blank. Whoever maintains the spreadsheet simply did not fill it in for every line.

PATTERN 4: CASE INCONSISTENCY

Unstandardized Casing

The same plant's line names appear in Title Case in one file, lowercase in another, and UPPERCASE in a third. This reflects the reality that the spreadsheet is typed into by different people at different plants over different years, with no shared style guide.

Trustworthiness Implication

Master Data is supposed to be the tie-breaker — the "official" name to trust when ERP and MES disagree. This dataset makes the point that in real organizations, the tie-breaker is often *less* reliable than the systems it is supposed to arbitrate between, not more.

8 QMS — Quality & Non-Conformance

The QMS file represents the quality system every automotive engineer recognizes: when a part fails to meet specification, it is logged as a Non-Conformance Report (NCR) with what was wrong, where it was caught, and what happened to the part.

Field Mapping Across Schema Variants

CONCEPT	VARIANT A (PAINT DE)	VARIANT B (BIW DE)	VARIANT C (ENGINE DE)	VARIANT D (ASSEMBLY IN)
NCR ID	QualityRecordID	QualityRecordID	NCRNumber	NCR_ID
Part	Part_No	Part_No	Component	PartNumber
Defect	Defect_Type	Defect_Type	FailureMode	DefectCode
Station	Station_Ref	Station_Ref	DetectionPoint	DetectedAtStation
Disposition	DispositionCode	DispositionCode	Disposition	Disposition
Defect rate	PPM	PPM	PPMRate	PPM_Rate
8D status	EightD_Stage	EightD_Stage	*absent*	EightD_Status

Table 8.1 — QMS field names across four schema variants. One variant (Engine DE) drops the 8D status field entirely.

Sample Data — Variant A (Paint Shop DE)

```
QualityRecordID,Part_No,Defect_Type,Station_Ref,DispositionCode,PPM,EightD_Stage NCR-DE-00001,MER-INTERIOR-7764-E,ASSY-CROSS-THREAD,PS-S01,Scrap,358,Closed NCR-DE-00002,MER-CHASSIS-2382-C,PAINT-ORANGE-PEEL,PS-S01,Scrap,301,Closed NCR-DE-00003,MER-BODY-4229-E,MATERIAL-CONTAMINATION,PS-S03,Rework,399,Closed
```

Sample Data — Variant C (Engine Plant DE, no 8D field)

```
NCRNumber,Component,FailureMode,DetectionPoint,Disposition,PPMRate NCR-DE-00001,MER-BODY-3110-D,ELEC-OPEN-CIRCUIT,EN-S01,Scrap,235 NCR-DE-00002,MER-BODY-8136-E,DIM-OUT-
```

```
OF-TOLERANCE,EN-S03,Scrap,388 NCR-DE-00003,MER-ELEC-3025-C,ASSY-WRONG-PART,EN-
S01,Rework,286
```

Industry-Specific Terms

> **PPM_Rate — Parts Per Million**
>
> The single most universal KPI in automotive quality. The value is formatted as a plain integer (e.g., 358, 99, 235) representing raw parts-per-million counts, not percentages. This formatting choice is deliberate: a percentage format would immediately signal synthetic data to anyone familiar with real QMS exports.

> **EightD_Status — 8-Step Problem Solving**
>
> The 8D methodology is an eight-step formal problem-solving process standard across the automotive industry. Status values are either `Closed` or `Open-D<n>` (indicating the process is currently stuck at step *n*). This is exactly the shorthand a real QMS system uses.

Disposition Distribution

Every NCR carries a disposition — the outcome for the defective part. The distribution is realistic: most defects are reworked because scrapping is expensive.

Figure 8.1 — Disposition Distribution Across the Network

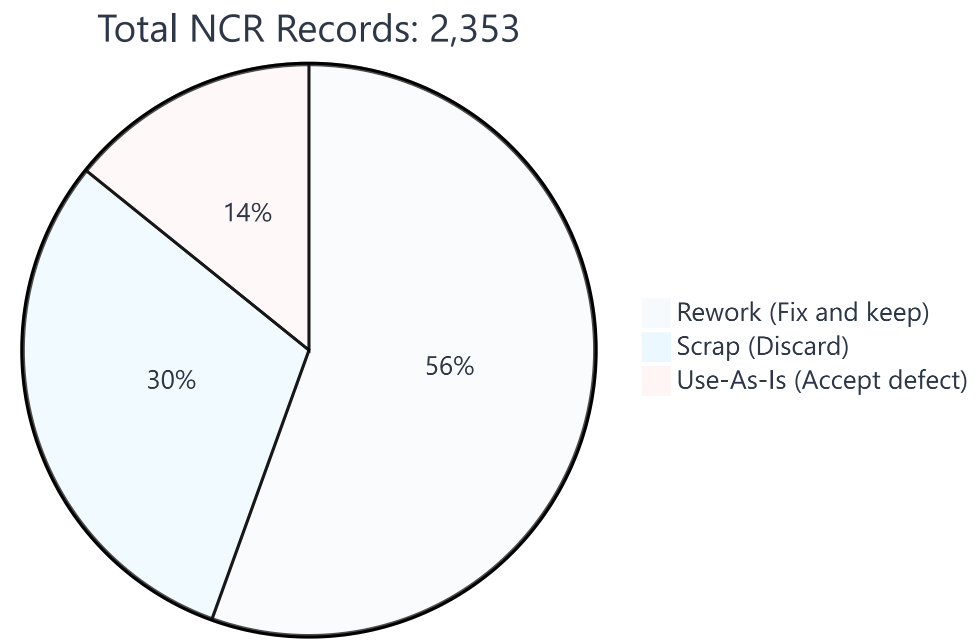

Figure 8.1 — Disposition distribution across all QMS files. The skew toward Rework reflects real-world economics: scrapping is expensive, so most defects are repaired.

Inherited Identity Problem

The `DetectedAtStation` field carries its own quiet identity challenge. Values like `PS-S01`, `BIW-S01`, and `EN-S01` each use a plant-abbreviation prefix plus a locally-numbered station. That local station number follows the same `S01`, `S02`, `S03` numbering scheme that MES uses, with the identical collision problem: `S01` at Paint Shop and `S01` at Engine Plant DE are two totally different physical stations.

9 EAM — Maintenance Management

The EAM file represents IBM Maximo or SAP PM-style maintenance management. Every service event — routine, breakdown, or emergency — generates a work order here. The same physical machines from ERP reappear, but under completely different identifier formats.

Field Mapping Across Schema Variants

CONCEPT	VARIANT A (ENGINE DE)	VARIANT B (STAMPING DE)	VARIANT C (PAINT DE)
Work order ID	WorkOrderID	OrderRef	WorkOrderID
Equipment	EquipmentID	EquipID	EquipmentID
Type	WOType	Type	WOType
Failure code	FailureCode	*absent*	FailureCode
Downtime	DowntimeMinutes	DowntimeMinutes	DowntimeMinutes
Reliability	MTBF_Hours	MTBFHours	MTBF_Hours

Table 9.1 — EAM field names across three schema variants. One variant (Stamping DE) drops the failure-code field entirely.

Sample Data — Variant A (Engine Plant DE)

```
WorkOrderID,EquipmentID,WOType,FailureCode,DowntimeMinutes,MTBF_Hours WO-2010001,ASSET-00001,Preventive,,0, WO-2010002,ASSET-00001,Preventive,,0, WO-2010003,ASSET-00001,Corrective,CALIBRATION-DRIFT,222,978 WO-2010004,ASSET-00001,Emergency,HYDRAULIC-LEAK,100,965 WO-2010005,ASSET-00001,Preventive,,0,
```

The Same Machine, Two Different Names

Compare the exact same physical machine across two systems:

ERP View

```
Werk: P1000 Anlagennummer: 000010000001 Description: Torque Application Unit
Manufacturer: DMG MORI
```

EAM View

```
WorkOrderID: WO-2010001 EquipmentID: ASSET-00001 Type: Preventive
```

Same machine. Two totally unrelated ID formats. ERP uses a 12-digit zero-padded number (`000010000001`); EAM uses an `ASSET-` prefix format (`ASSET-00001`). There is no shared key connecting these two rows. Reconciling them is precisely the job the downstream mapping system must perform.

Maintenance Type Distribution

Preventive maintenance rows are deliberately sparse — blank failure code, zero downtime, blank MTBF — because routine maintenance is not a failure. Only Corrective and Emergency rows carry real failure codes, real downtime minutes, and real MTBF values.

TYPE	NETWORK SHARE	FAILURE CODE	DOWNTIME	MTBF
PREVENTIVE	~55%	Blank	0	Blank
CORRECTIVE	~35%	Populated	Real minutes	Real hours
EMERGENCY	~10%	Populated	Real minutes	Real hours

Table 9.2 — Maintenance type distribution and field population rules. Preventive rows are sparse by design; only unplanned events carry failure metadata.

10 WMS — Material Flow & Shipments

The WMS file represents Manhattan or SAP EWM-style warehouse management, tracking Kanban-based material flow — what shipped from which plant to which plant, and in what quantity. Unlike the previous systems, every row is inherently a relationship between two plants.

Field Mapping Across Schema Variants

CONCEPT	VARIANT A (STAMPING DE)	VARIANT B (ASSEMBLY DE)	VARIANT C (ENGINE DE)
Material	PartNo	PartNo	MaterialNumber
Storage location	Bin	Bin	StorageBin
Quantity	OnHandQty	OnHandQty	QtyOnHand
Kanban card	*absent*	*absent*	KanbanCardID
Sender	IssuingPlant	IssuingPlant	IssuingPlant
Receiver	ReceivingPlant	ReceivingPlant	ReceivingPlant

Table 10.1 — WMS field names across three schema variants. Only Variant C includes the KanbanCardID field.

Sample Data — Variant A (Stamping Plant DE → BIW Plant DE)

```
PartNo,Bin,OnHandQty,IssuingPlant,ReceivingPlant MAT-216928,F-11-1,4852,Stamping
Plant DE,BIW/Welding Plant DE MAT-952257,E-29-5,2719,Stamping Plant DE,BIW/Welding
Plant DE MAT-997028,E-16-1,2288,Stamping Plant DE,BIW/Welding Plant DE
```

Sample Data — Variant B (Battery Plant PL → Assembly Plant DE)

```
PartNo,Bin,OnHandQty,IssuingPlant,ReceivingPlant MAT-321465,F-02-5,1741,Battery &
eDrive Plant PL,Assembly Plant DE MAT-254863,B-10-5,2802,Battery & eDrive Plant
PL,Assembly Plant DE MAT-482201,D-23-1,4960,Battery & eDrive Plant PL,Assembly Plant
DE
```

Supply Chain Verification

Every row's `IssuingPlant` / `ReceivingPlant` pair is generated to match a real edge in the supply chain — nothing is random. This allows direct verification of the network topology from Phase 2.

Figure 10.1 — Verified Supply Lanes in WMS Data

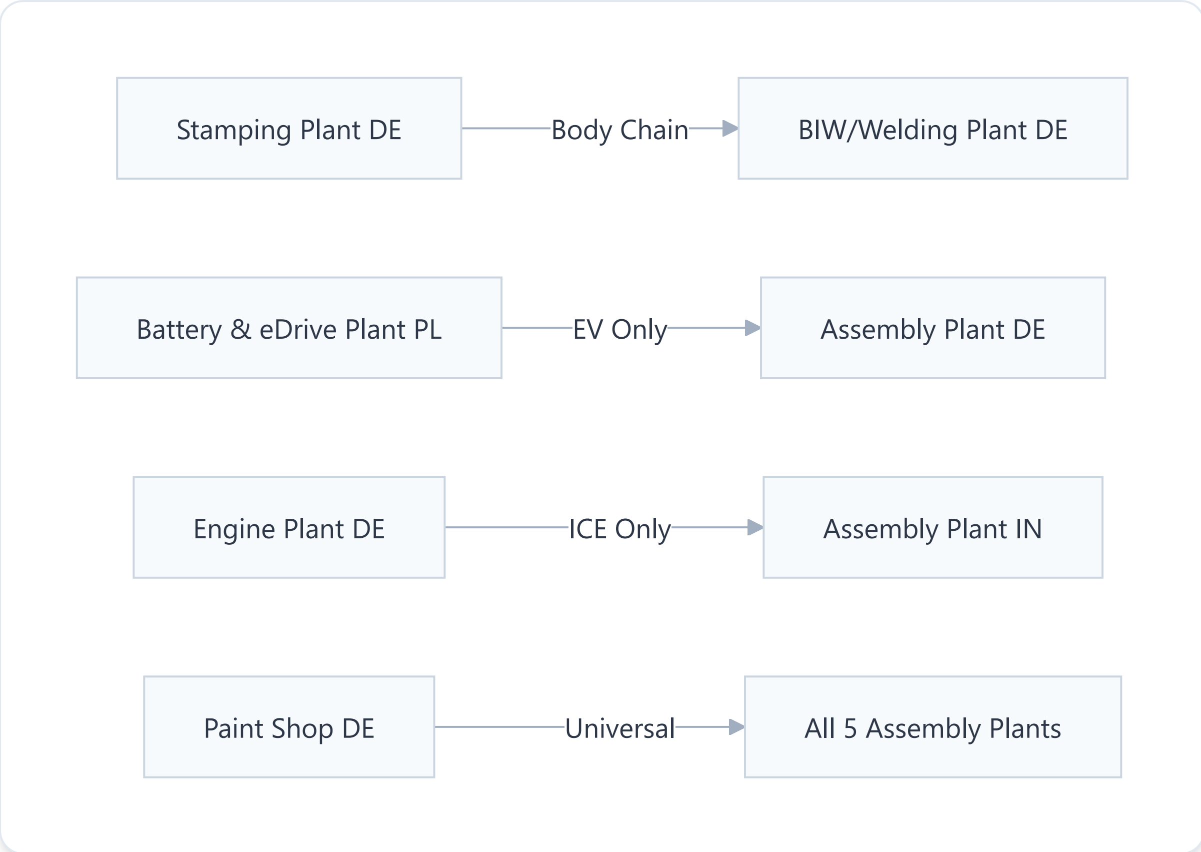

Figure 10.1 — Verified supply lanes in WMS data. The shape of who ships to whom is guaranteed correct by construction; the exact quantities are synthetic.

> **Kanban Context**
>
> KanbanCardID references the lean manufacturing / Toyota Production System concept of a physical or digital card that authorizes replenishment. Kanban is near-universal across automotive supply chains specifically, which is why it appears here even though a generic WMS might not include it.

11 SPC — Statistical Process Control

The SPC file represents Minitab or Q-DAS-style statistical process control — repeated measurements of a specific product characteristic (a dimension, a torque value, or any controlled parameter), checked against specification limits over time. This is the raw material behind a control chart.

Field Mapping Across Schema Variants

CONCEPT	VARIANT A (SEATING PL)	VARIANT B (STAMPING DE)	VARIANT C (ENGINE DE)	VARIANT D (DROPS CPK)
Characteristic	Characteristic_No	Characteristic_No	Characteristic_No	CharID
Station	Station_Ref	Station_Ref	Station_Ref	Station
Measured value	Value	Value	Value	Reading
Upper spec limit	UpperSpecLimit	UpperSpecLimit	UpperSpecLimit	USL
Lower spec limit	LowerSpecLimit	LowerSpecLimit	LowerSpecLimit	LSL
Process capability	ProcessCapability	ProcessCapability	ProcessCapability	*absent*

Table 11.1 — SPC field names across four schema variants. One variant drops the Cpk (process capability) field entirely.

Sample Data — Variant A (Seating & Interior Plant PL)

```
Characteristic_No,Station_Ref,Value,UpperSpecLimit,LowerSpecLimit,ProcessCapability
K1001,S01,154.787,157.362,152.145,1.68 K1001,S01,154.463,157.362,152.145,1.68
K1001,S01,155.001,157.362,152.145,1.68 K1002,S01,184.253,185.854,181.702,0.83
K1002,S01,183.368,185.854,181.702,0.83
```

Cpk — Process Capability Index

Cpk is not a random number. It is mechanically tied to how tightly the measurements cluster relative to the specification limits.

Capable Process (Cpk = 1.68)

```
Value: 154.787 USL: 157.362 LSL: 152.145 Cpk: 1.68
```

Three readings cluster tightly near the center of the 152–157 window. Industry convention considers Cpk above ~1.33 as capable.

Marginal Process (Cpk = 0.83)

```
Value: 184.253 USL: 185.854 LSL: 181.702 Cpk: 0.83
```

Readings spread out much more relative to the spec window. A Cpk below 1.0 indicates a marginal or borderline process.

The correlation between measurement spread and Cpk is deliberate. A dataset that placed random numbers in the Cpk column would fail scrutiny from anyone who has read a real control chart. You will also see the same `CharacteristicID` repeated across consecutive rows with slightly different `MeasuredValues` — this is intentional, as a real SPC system logs repeated readings of the same characteristic over time.

Honest Gap: SPC Absent at Tier-2 Plants

Deliberate Absence

SPC files are completely absent at two plants:

- **Solara Metals Coil Plant (DE)** — no spc.csv
- **Ferrovia Harness Components Plant (VN)** — no spc.csv

This is not a bug or oversight. These are the network's two smallest Tier-2 raw-material suppliers. Small, less-mature suppliers often genuinely do not have SPC tooling deployed. The absence itself is a realistic data point.

12 PLM & APS/SCM — Network-Level Systems

These two systems work differently from everything covered so far. They do not describe one plant; they describe the entire company at once. There are only 4 files total: 3 for PLM (one per vehicle platform) and 1 for APS/SCM (the entire network).

PLM — Product Structure & Bill of Materials

The PLM file is a real tree, not a flat list — that is the entire purpose of a Bill of Materials. Every part traces back to the vehicle through its parent.

Sample Data — Compact Platform BOM (Tree Structure)

```
ItemNumber,Revision,ParentItem,ChangeNotice,EffectiveDate MER-VEH-COMPACT-0001,C,,ECN-2022-0028,2021-05-12 MER-BODY-0002,A,MER-VEH-COMPACT-0001,ECN-2023-0830,2025-12-14 MER-BODY-SHELL-0003,A,MER-BODY-0002,,2023-09-25 MER-DOORS-0004,C,MER-BODY-0002,ECN-2021-0327,2025-04-09 MER-HOOD-0005,A,MER-BODY-0002,,2021-12-25 MER-CHASSIS-0009,A,MER-VEH-COMPACT-0001,ECN-2021-0503,2022-04-28 MER-FRONT-SUSPENSION-0010,A,MER-CHASSIS-0009,,2021-05-15
```

Figure 12.1 — Compact Platform BOM Hierarchy

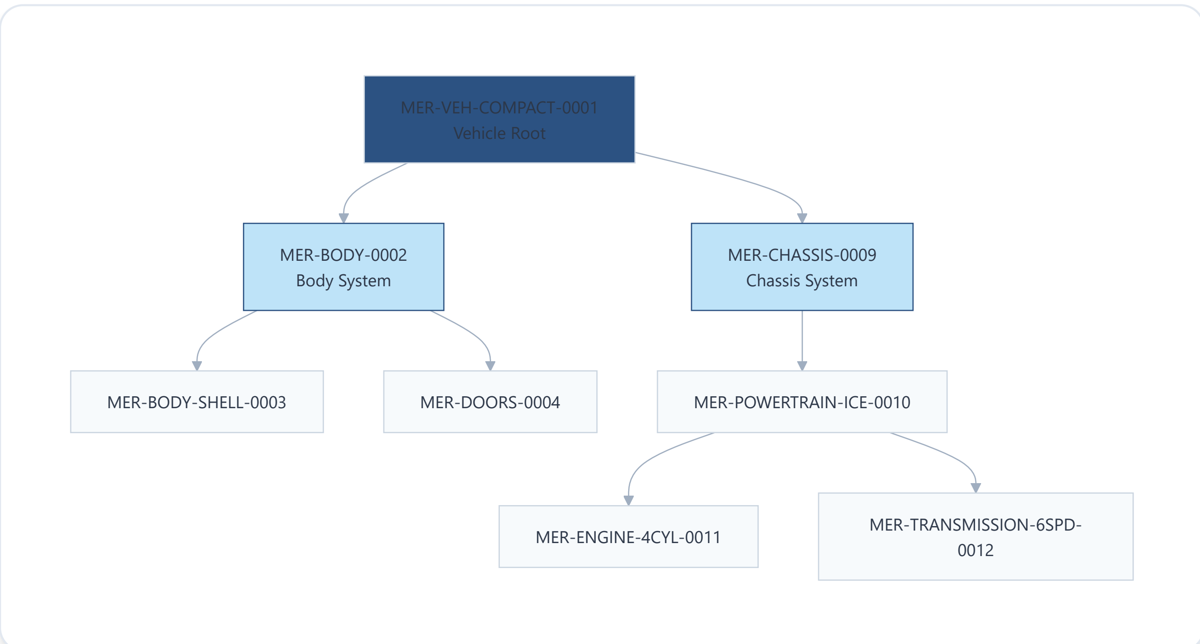

Figure 12.1 — Compact platform BOM hierarchy. The tree structure traces every part back to the vehicle root. Platform-specific parts (engine vs. battery) appear only in their respective platform files.

APS/SCM — Capacity Planning

APS/SCM is inherently cross-plant. The single network-level file contains weekly capacity plans for every plant.

Sample Data

```
PlanningPeriod,PlantID,PlannedVolume,CapacityHours,SupplierLeadTimeDays 2020-W02,PL-013,1766,2130,43 2020-W03,PL-013,1645,2935,39 2020-W04,PL-013,2205,2790,51 2020-W02,PL-011,16740,499,6 2020-W03,PL-011,19384,409,27 2020-W04,PL-011,5889,508,20
```

> **Assembly Plant DE (PL-013)**
>
> **PlannedVolume:** ~1,600–2,200 units/week
>
> **CapacityHours:** ~2,100–2,900 hours/week
>
> **Profile:** Lower volume of high-value finished vehicles with substantial capacity behind each unit.

> **Solara Metals (PL-011, Tier-2)**
>
> **PlannedVolume:** ~5,900–19,400 units/week
>
> **CapacityHours:** ~400–500 hours/week
>
> **Profile:** Much higher volume of a simpler, lower-value raw material product through a smaller facility.

The volume contrast between a final assembly plant and a Tier-2 raw-material supplier is the right shape, not a coincidence. If these two rows looked similar, that would be the signal that the data was not grounded in real operational logic.

13 Messiness Taxonomy (M1–M4)

The dataset injects four categories of deliberate disagreement, each calibrated against real-world data-quality research. These are not random errors — they are the specific, well-known categories of disagreement that enterprise systems exhibit when they were never designed to interoperate.

Figure 13.1 — The Four Messiness Categories

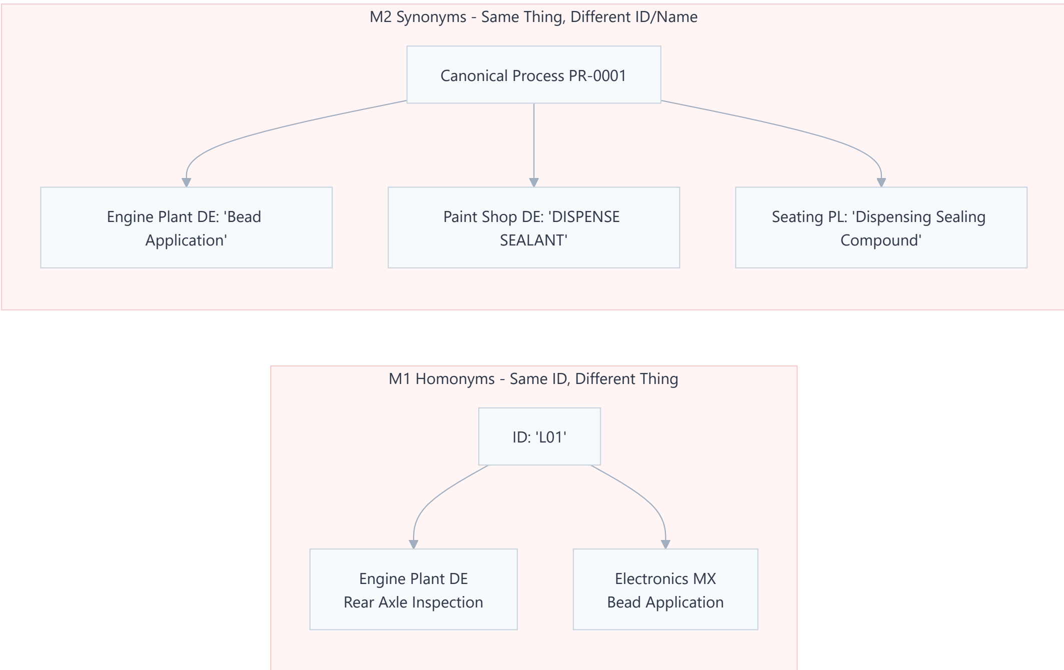

Figure 13.1 — The four categories of deliberate data-quality disagreement injected across the dataset. Each category is calibrated against real-world enterprise data-quality research.

Calibration Against Real-World Research

The injection rates are not arbitrary. They were calibrated against published figures from an actual Bosch case study on multi-system identity reconciliation in automotive manufacturing.

CATEGORY	INJECTED RATE	TARGET / CALIBRATION SOURCE	STATUS
M1 Homonym	136 colliding plant pairs	Target: ≥3 plant pairs	EXCEEDED
M2 Synonym	138 of 200 processes (69.0%)	Target range: 65–70%	WITHIN RANGE
M3 Missing Item	5253 links, 452 unlinked (8.6%)	Target range: 7–9%	WITHIN RANGE
M4 Cross-Tier	104 colliding plant pairs	Target: ≥2 confirmed cross-tier collisions	EXCEEDED

Table 13.1 — Messiness injection metrics calibrated against real-world automotive data-quality research. All categories meet or exceed their targets.

Validation Principle

When evaluating any file from this dataset, apparent inconsistencies should be checked against these four categories before being flagged as errors:

1. **Missing/blank field** → Expected under M3 or general realism (~3% ERP, ~20% MD).
2. **Same-looking ID meaning different things** → Expected under M1 (homonym) or M4 (cross-tier).
3. **Same real thing spelled differently** → Expected under M2 (synonym) or general case inconsistency.
4. **Machine/record that does not connect to anything** → Expected under M3 (missing item), such as the 452 explicitly unlinked machines.

If an inconsistency does not fit one of these four buckets, it warrants further investigation as a potential genuine gap rather than intended realism.

14 Reference Files — Internal Answer Key

Two JSON files reside in the `reference\` folder. These files are the internal grading infrastructure, not part of the demo data. They should never be exposed during validation exercises.

generation_manifest.json — The "Did It Work" Report

This file is not data about vehicles or plants — it is metadata about the dataset itself. It records the generation seed and, most importantly, tells you exactly how much deliberate messiness actually landed, checked against its own targets.


```
{ "seed": 42, "network": { "companies": 5, "plants": 17, "platforms": 3 },
"sic_injection": { "M1_homonym": { "colliding_plant_pairs": 136, "target": ">=3 plant
pairs" }, "M2_synonym": { "processes_with_variants": 138, "rate_pct": 69.0,
"target_pct_range": [65, 70], "total_processes": 200 }, "M3_missing_item": {
"linked_machine_count": 5253, "rate_pct": 8.6, "target_pct_range": [7, 9],
"unlinkable_count": 452 }, "M4_cross_tier": { "colliding_plant_pairs": 104, "target":
">=2 confirmed cross-tier collisions" } } }
```


This manifest serves as the receipt for realism. If the question arises whether the amount of messiness is realistic or arbitrary, this file provides the calibrated answer: every rate was set against real published figures from actual industry research.

ground_truth_reference.json — The Complete Decoder

This is the answer key. It knows, with certainty, what every messy ID in every CSV really refers to. Three illustrative entries:


```
// Line ID resolution: "L01" at Engine Plant DE { "global_key":
"AUTO_mother_Plant1_Sys1_Line1", "local_line_id": "L01", "plant": "Engine Plant DE",
"tier_role": "mother" } // Process PR-0001: every plant's local name in one place {
"process_number": "PR-0001", "canonical_name": "Adhesive Dispensing",
"plant_local_names": { "Engine Plant DE": "Bead Application", "Paint Shop DE":
"DISPENSE SEALANT", "Assembly Plant CN": "Dispense Sealant", "Solara Metals Coil
Plant": "Bead Application", // ... 12 more plant entries } } // Explicitly unlinked
machine (M3 missing item) { "machine_global_key": "AUTO_Plant1_Machine1", "plant":
"Engine Plant DE", "status": "not_connected", "work_unit_global_key":
"AUTO_mother_Plant1_WU1" }
```

ENTITY TYPE	COUNT IN GROUND TRUTH	PURPOSE
Lines	875	Resolves every local LineID to a unique global key per plant
Stations	2,628	Maps local station numbers to physical locations
Processes	200	Records every plant's local name for each canonical process
Machine–Work Unit Links	5,253	Tracks which machines are connected (linked) vs. deliberately unlinked

Table 14.1 — Ground truth entity counts and their resolution purpose.

Critical Isolation Rule

The entire premise of the dataset is that each export must be evaluated on its own terms — "does this file, in isolation, look like something a real system produced?" The moment the ground truth is exposed, evaluation ceases to be about the data's standalone realism and becomes about reading the solution. Keep `raw\automotive\` and `reference\` strictly separated.